

Lidar: How laser pulses shape the future of autonomous vehicles and remote sensing

Lidar technology is a pivotal innovation in autonomous vehicles and remote sensing. By harnessing the power of laser pulses, lidar measures distances and pinpoints objects with impressive accuracy. Prioritising safety, the automotive industry has eschewed traditional lasers that could pose risks to the eyes, instead opting for ‘visually safe lasers’ in lidar configurations. Lidar technology fundamentally relies on laser pulses to detect surrounding objects. The ‘time of flight’ measurement is central to its operation, calculating the time a laser pulse takes to strike an object and reach back.

This mechanism facilitates an accurate estimate of an object’s distance. By gauging the reflectivity of the returned signal, one can infer specific characteristics of the detected object, like its colour or whether it has a metallic composition. Advanced software algorithms process and classify the data the lidar collects, effectively distinguishing between various entities—for example, differentiating a tree from a pedestrian or a car.

Elements of lidar testing

To ensure that a lidar system is up to the mark, the device under test (DUT) undergoes rigorous testing. It includes:

Hardware testing. In every lidar device’s heart lies a symphony of finely tuned optical and mechanical systems. Their precision is achieved through rigorous calibration, ensuring peak performance right out of the box. But that is just the beginning. Each device then undergoes a rigorous performance sensitivity analysis, a deep dive into how it responds under various conditions, assessing its reliability at each step.

The pièce de résistance? An exhaustive accuracy assessment that zeroes in on the lidar’s ability to measure distance and reflectivity. This capstone test is designed to guarantee that the information lidar provides is not only accurate but remains steadfastly consistent, no matter the environment.

Software testing. No stone is left unturned regarding the intricate world of lidar software testing. The crucial evaluation of detection algorithms is at the forefront of this rigorous process. Expert eyes meticulously gauge these algorithms for their precision and flair in identifying targets. The nuanced art of object classification is put under the microscope, ensuring an unerring ability to differentiate between a pedestrian, a zooming vehicle, or nature’s obstacles. But there’s more. Central to the process is the system’s talent for target reconstruction. It’s not just about pinpoint accuracy; it’s about the lightning-quick speed of this reconstruction.

Challenges of testing lidar systems within manufacturing

Lidar testing is no small feat, and central to this challenge is the immense spatial demand. Imagine this: target boards placed everywhere, from a mere 5 metres away to distant spots almost touching the 300-metre mark. This expansive setup consumes vast floor spaces, especially when evaluations stretch over extensive ranges. But it is not just about space. The waters get muddier with non-standard testing procedures. The absence of a uniform standard can inject unwanted variability, casting a shadow over result reliability. And if you think that’s all, consider the clock. Testing is not just intricate, but it’s time-consuming. The constant need to gauge lidar efficiency at varying distances means frequent setup

reshuffles, extending the timeline further.

In the dynamic landscape of lidar technology, refining testing methods to scale new heights of efficiency is a constant tug-of-war. Remember the days when testing lidar meant sprawling setups across great distances? Such traditional methods have always been a logistical puzzle, demanding time and space in spades. Yet, in this maze, there is a glimmer of streamlined processes. Adrian says, “With the prowess of big data analytics, pufferfish manufacturing analytics emerges not just as another software but as an insightful tool.” It gives manufacturers a microscopic view into component quality, ensuring consistency across parts from diverse suppliers. It’s an era where big data isn’t just about volume; it’s about refining quality assurance with surgical precision.

In a world where sustainability and safety converge, autonomous driving emerges as a beacon of innovation, fortified by relentless global support. This transformative journey is anchored by two technological titans: radars and lidars. The industry’s pulse is palpable in the meticulous craft of radar systems and the intricate applications of lidar. But, as with all pioneering ventures, roadblocks are inevitable. Lidar testing, alongside the pursuit of universal standards, poses significant challenges. Yet, in this intricate dance of technology and data analytics, the industry is gearing up to convert these challenges into monumental leaps forward. It is a riveting era, steering the automotive domain towards an electrified, self-directed, and luminous horizon for humanity. **EFY**

This article is from a tech talk session at EFY Expo 2023, Pune, by Adrian Gaudencio Ababa, Business Development Manager - Automotive and Energy Solutions, Keysight Singapore. Transcribed and curated by Akanksha Sondhi Gaur, Research Analyst and Journalist at EFY.